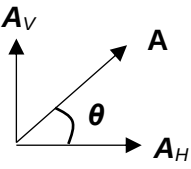


2	(a)	Moment of a force is the <u>product of the force and the perpendicular distance to a pivot.</u>	B1
		Marker's comment Students need to learn the definition and follow the definition closely. Simply saying that it is the product of force and perpendicular distance is not sufficient. Students need to state clearly the perpendicular distance is from the line of action of the force to the pivot. Saying "perpendicular distance from the pivot" is not clear enough. There was also some students who mentioned "perpendicular distance from the point the force acts to the pivot" which was incorrect.	
	(b)	(i) Taking moments about A, $T(2.8) = 500(1.4 \sin 30^\circ) + 4000(2.8 \sin 30^\circ)$ $T = 2125 \text{ N}$ $T = 2.1 \text{ kN (shown)}$	M1 A0
		Marker's comment Most students were able to show the tension by taking moments about point A. There were a handful who were unable to find moment of forces correctly.	
		(ii) $T_v = T \cos 60^\circ$ $= 2100 \cos 60^\circ$ $= 1050 \text{ N}$	A1
		Marker's comment Most students were able to resolve the tension correctly.	
		(iii) There is an <u>upward vertical component of force acting on the beam at A</u> such that upward force at A + T_v = sum of downward forces of weight of beam and load.	B1
		Marker's comment Not well done. The key idea that there is an "upward vertical component of the force acting on the beam" was not clearly brought across. Also, there were several students who incorrectly used the term "normal contact force" from the wall. Normal contact force acts perpendicular to the vertical wall and thus would have been acting horizontally and hence not awarded credit.	

		(iv) <u>Vertical equilibrium:</u> $A_v + T_v = 500 + 4000$ $A_v = 4500 - 1050$ $= 3450 \text{ N}$ <u>Horizontal equilibrium:</u> $A_H = T \sin 60 = 2100 \sin 60$ $= 1820 \text{ N}$ Hence magnitude of force at A = $(1820^2 + 3450^2)^{1/2}$ $= 3900 \text{ N}$  $\tan \theta = 3450 / 1820$ $\theta = 62^\circ$ Direction is 62° above the positive horizontal.	C1 C1 A1 A1
		Marker's comment Poorly done. Many students were not able to recall the steps to solve this kind of question. The angle was poorly expressed. Students should draw diagram to show clearly the angle they were determining. Use of scale diagram to solve this question is not encouraged.	

3	(a)	Work done is the product of the force and the displacement moved in the direction of	P1
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